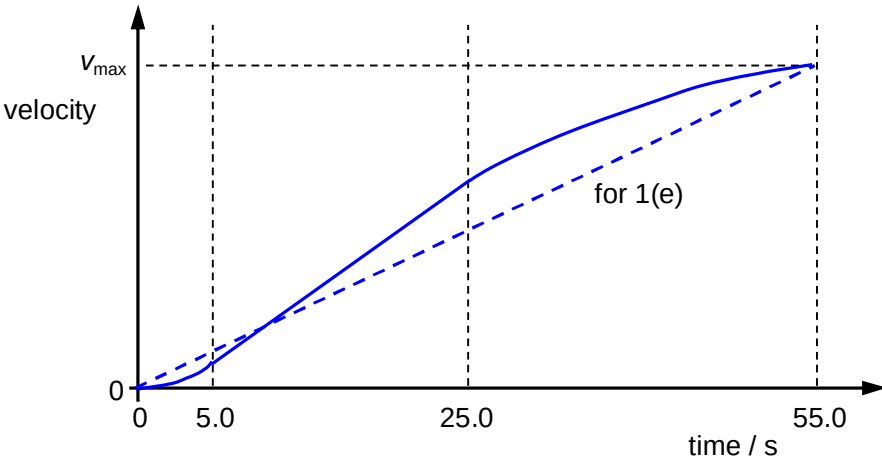


No.	Solution	Remarks
1(a)	Any two: Air resistance varies with speed of aeroplane. Friction runway exerts on aeroplane varies. Mass of fuel in aeroplane decreases during take-off.	[2] 1 mark for each correct answer
1(b)	By Newton's 2nd law, Change in momentum of the aeroplane = area under the graph = final momentum – initial momentum = final momentum at take-off - 0 Momentum $= \left(\frac{1}{2} \times 5.0 \times 1.13 \times 10^5 \right) + \left(20.0 \times 1.13 \times 10^5 \right) + \left[\frac{1}{2} (1.13 + 0.97) \times 10^5 \times 30.0 \right]$ $= 5.693 \times 10^6$ $\approx 5.69 \times 10^6 \text{ N s}$ <i>Comment:</i> - state the correct physics principle, as change in momentum is area under the force-time graph.	[1] for using area under graph [1] for correct expression and numerical substitution [1] for correct answer
1(c)	Using $p = mv$, $v_{\max} = \frac{p}{m}$ $= \frac{5.693 \times 10^6}{7.45 \times 10^4}$ $= 76.42$ $\approx 76.4 \text{ m s}^{-1}$	[1] for correct answer
1(d)		

No.	Solution	Remarks
	 Note: [1] Curve of increasing gradient from origin and line of constant gradient between 5.0 and 25.0 s. [1] Line of decreasing positive gradient after 25.0 s. <i>Comment:</i> - many sketched the graph with obvious discontinuity or kinks.	
1(e)	Distance = area under graph $= \frac{1}{2} \times 76.42 \times 55.0$ $= 2101$ $\approx 2200 \text{ m (Method is underestimation, refer to Fig. 3.2)}$ Accept 2000 – 3000 m <i>Comments:</i> - many resorted to kinematics calculation, based on constant acceleration without clear explanation. - the estimation principle using graphs poorly understood and applied by many.	[1] for correct estimation method [1] for correct answer

No.	Solution	Remarks
2(a)		[1] horizontal